

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460691.869 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.044
Transit depth (Rp/Rs)^2	8e-07 +/- 0.00078 [%]
Orbital Inclination (inc)	80.02 +/- 2.87
Airmass coefficient 1 (a1)	1.49 +/- 0.8
Airmass coefficient 2 (a2)	-0.05 +/- 0.41
Scatter in the residuals of the lightcurve fit is	54.94 %
Best Comparison Star	None
Optimal Aperture	1.98
Optimal Annulus	3.28
Transit Duration (day)	0.0763 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.0 ± 0.044 vs 0.1178 (deviation 2.23σ)
Duration fit vs published	0.0763 d vs 0.125 d (deviation 6.33σ)
Mid-time O-C	1.8 min
Depth SNR	0.0
Residual scatter	54.94 %

Light curve

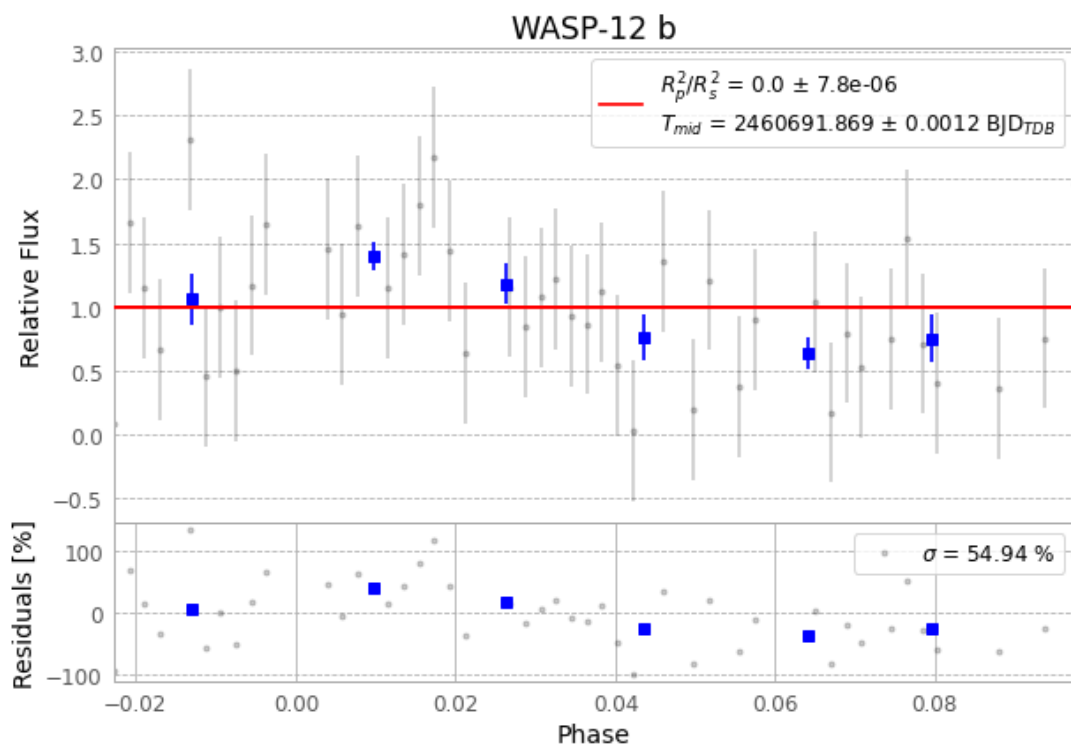


Figure 1 — FinalLightCurve_WASP-12 b_2025-01-16.png (SHA-256 b9cd15f2f998176b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [184, 67], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20ed30e65e1db181...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d

Data provenance

68 FITS frames (45 MB), WASP-12250116075416.FITS → WASP-12250116111516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250116.log (73b353054bcd...), FinalLightCurve_WASP-12 b_2025-01-16.png (b9cd15f2f998...), FinalLightCurve_WASP-12 b_2025-01-16.csv (beb7f46270d0...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:57Z.